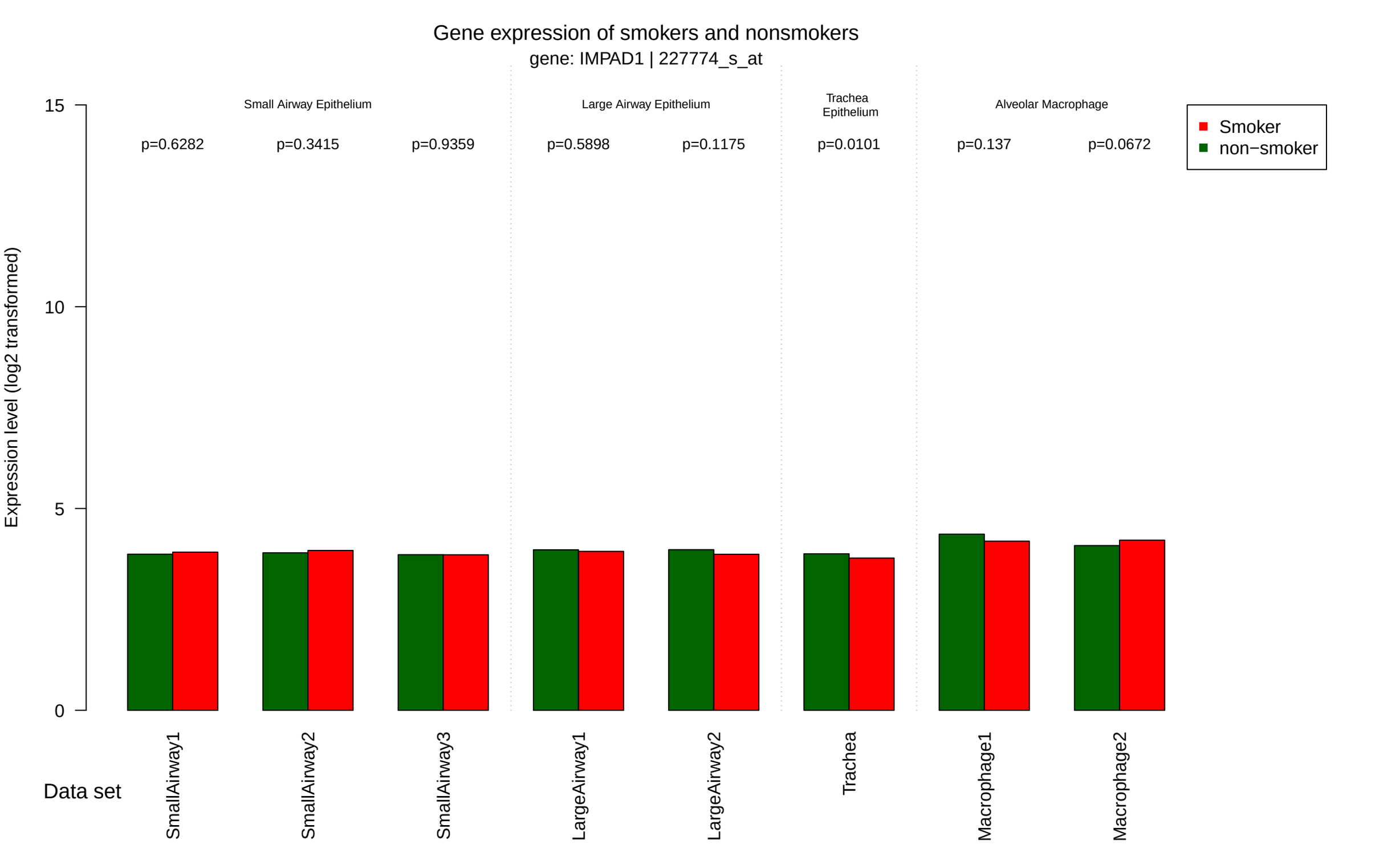
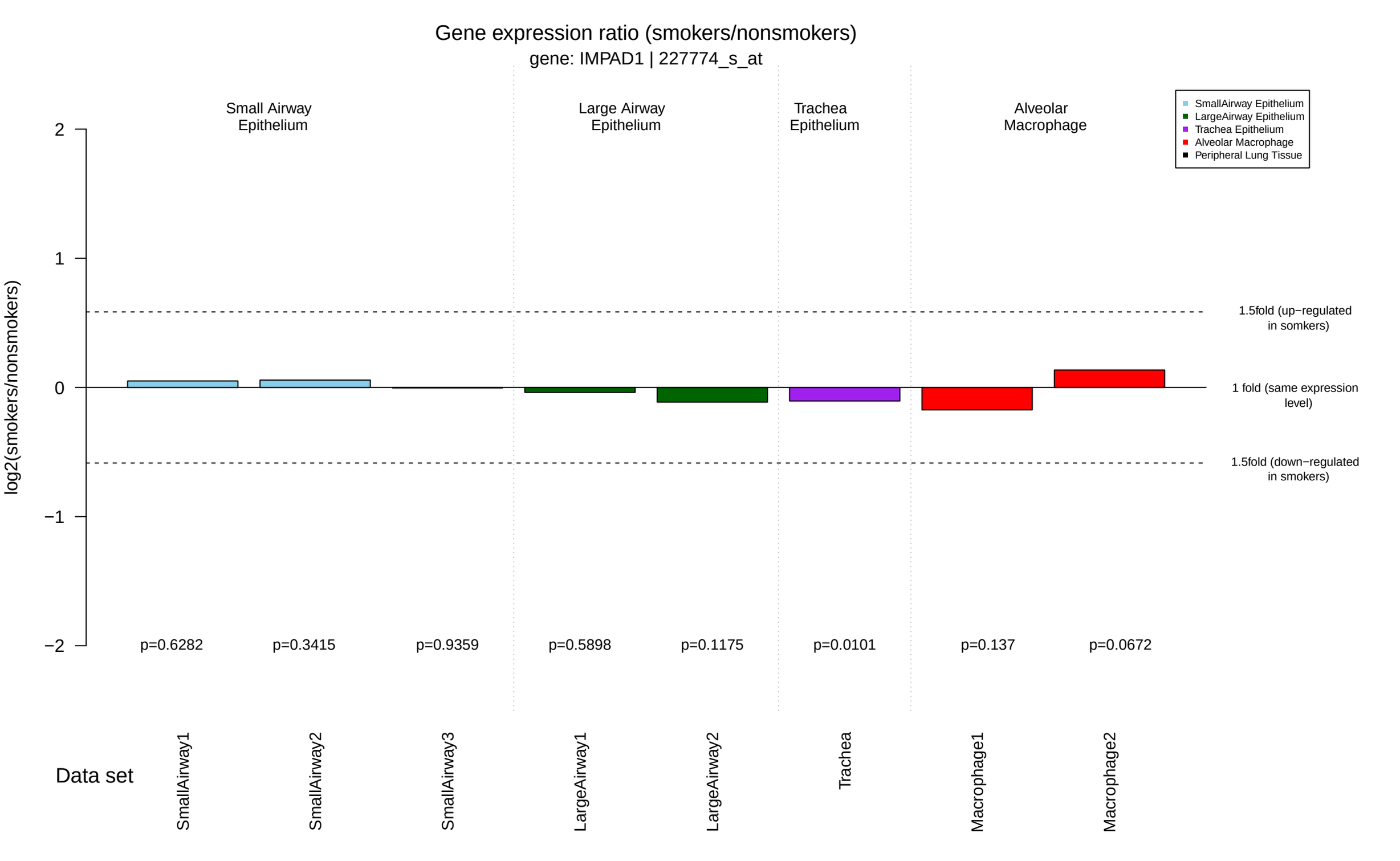
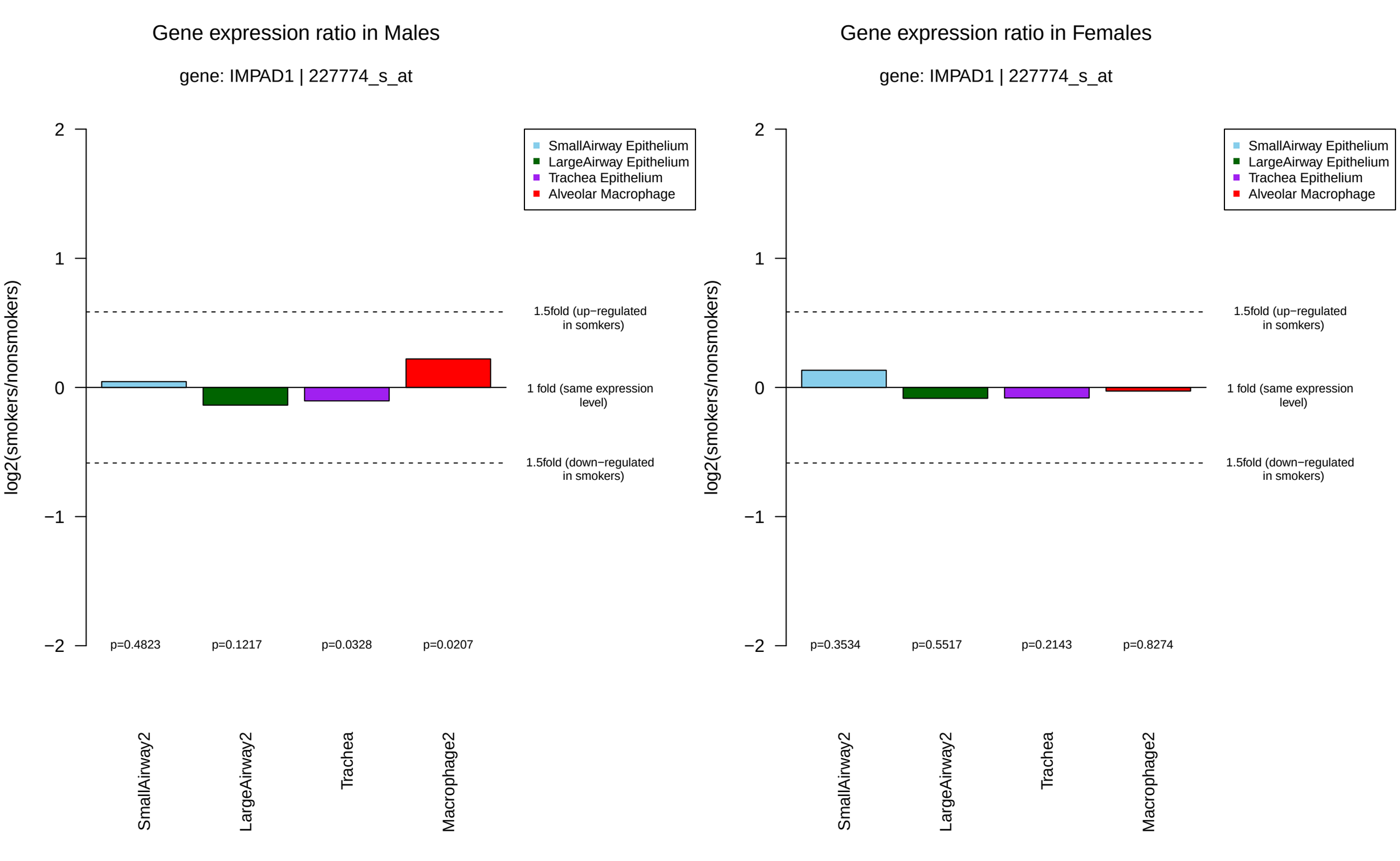
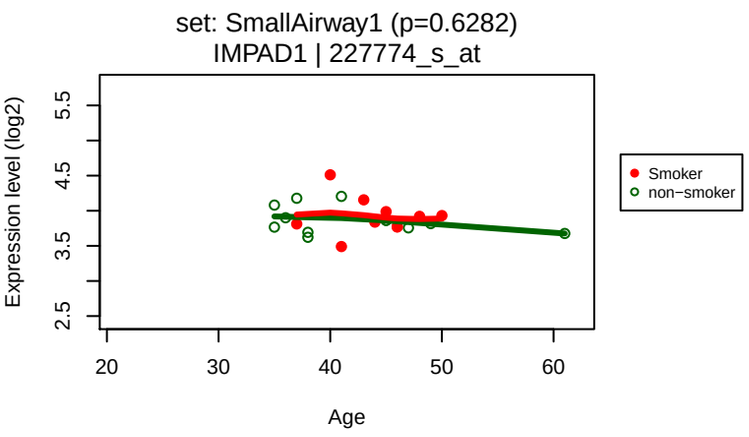
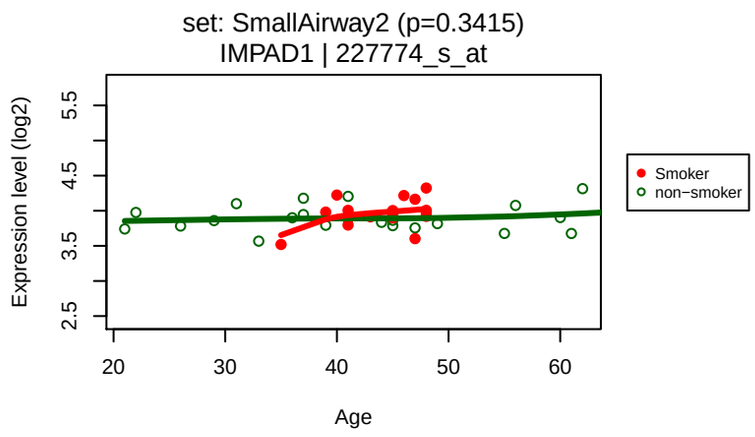
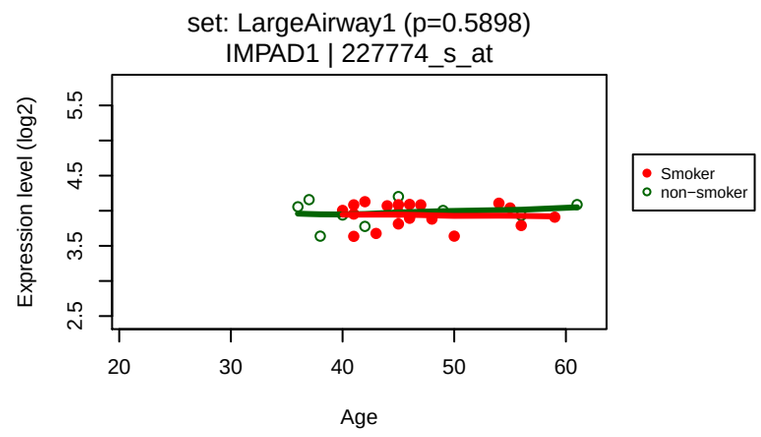

Gene expression of smokers and nonsmokers
gene: IMPAD1 | 227774_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6282
p=0.3415
p=0.9359
p=0.5898
p=0.1175
p=0.0101
p=0.137
p=0.0672
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

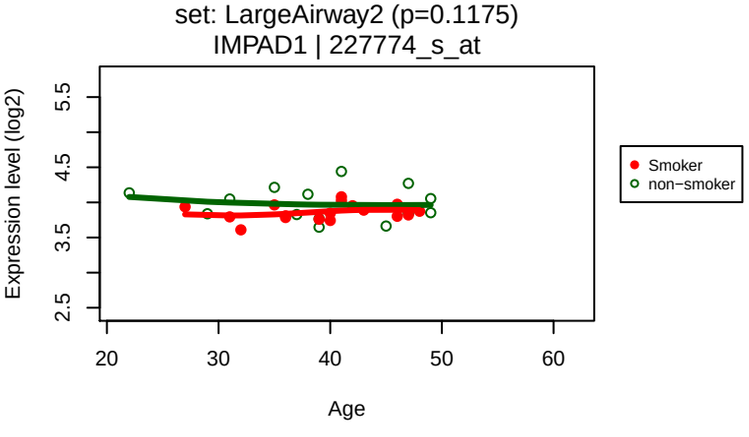

Gene expression ratio (smokers/nonsmokers)
gene: IMPAD1 | 227774_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6282
p=0.3415
p=0.9359
p=0.5898
p=0.1175
p=0.0101
p=0.137
p=0.0672
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

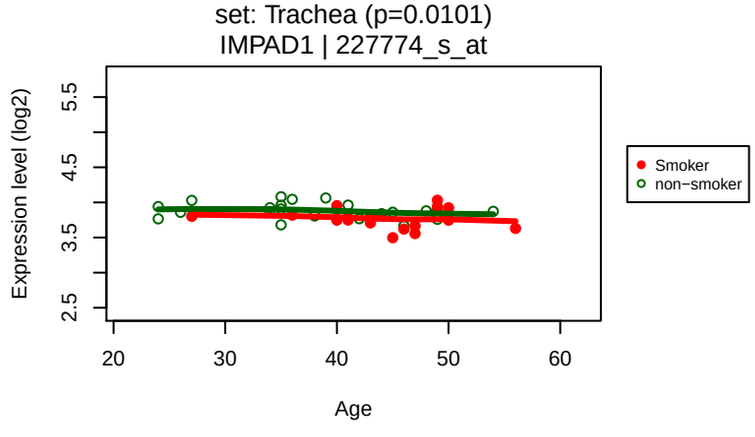

Gene expression ratio in Males
gene: IMPAD1 | 227774_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4823
p=0.1217
p=0.0328
p=0.0207
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IMPAD1 | 227774_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3534
p=0.5517
p=0.2143
p=0.8274
SmallAirway2
LargeAirway2
Trachea
Macrophage2

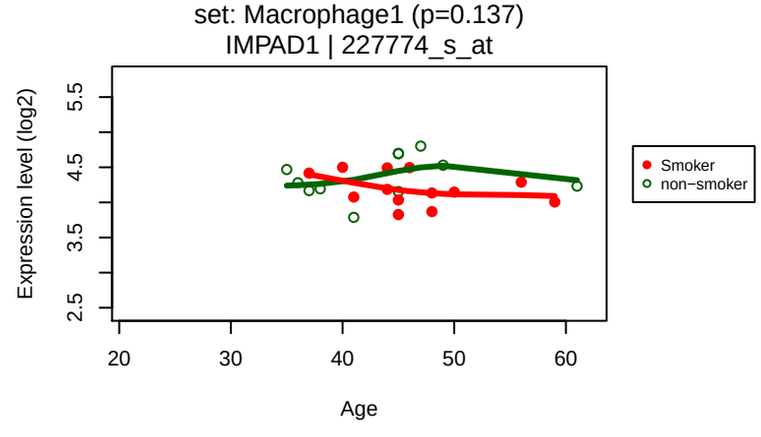

set: SmallAirway1 (p=0.6282)
IMPAD1 | 227774_s_at
Expression level (log2)
Age
Smoker
non−smoker

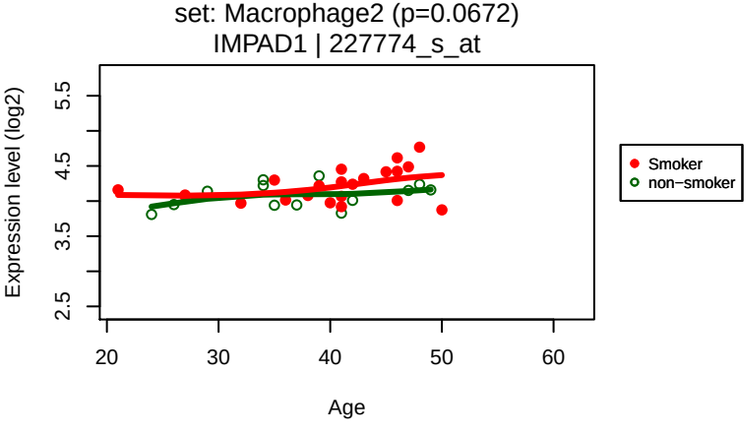

set: SmallAirway2 (p=0.3415)
IMPAD1 | 227774_s_at
Expression level (log2)
Age
Smoker
non−smoker

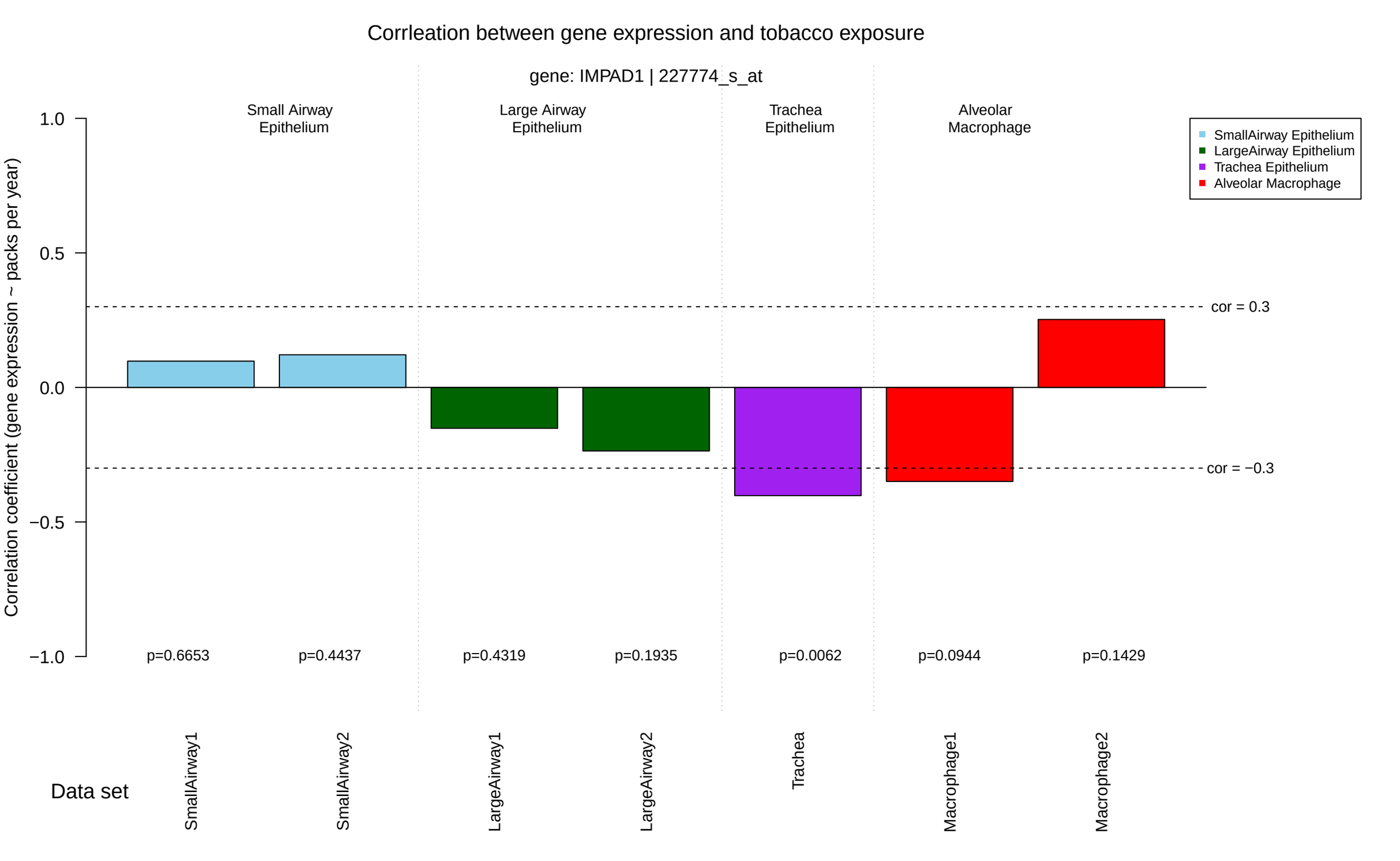

set: LargeAirway1 (p=0.5898)
IMPAD1 | 227774_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1175)
IMPAD1 | 227774_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0101)
IMPAD1 | 227774_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.137)
IMPAD1 | 227774_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0672)
IMPAD1 | 227774_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IMPAD1 | 227774_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6653
p=0.4437
p=0.4319
p=0.1935
p=0.0062
p=0.0944
p=0.1429
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2